LAPORAN PRAKTIKUM

TEKNOLOGI IOT II

Modul 6: Protokol Komunikasi Telemetri Industri (MQTT)

Disusun oleh :

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Kelas : TRI-3B

Dosen:

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Dr. Ir. Septafiansyah Dwi Putra

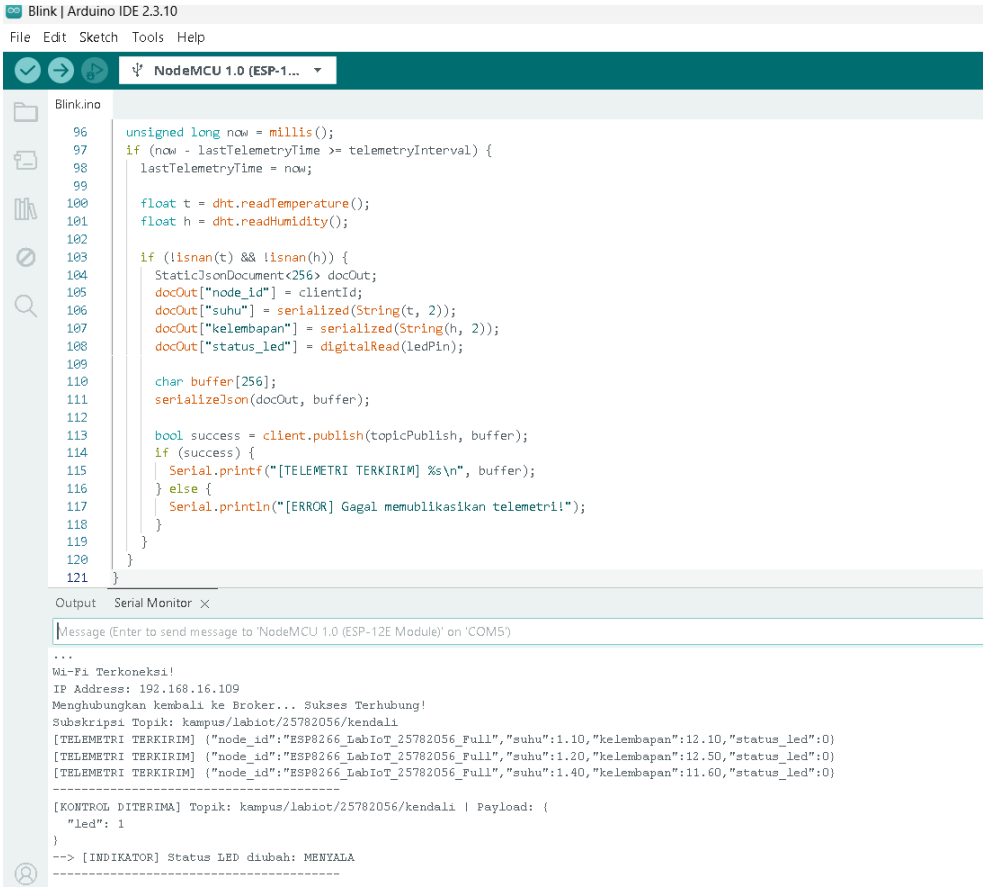
TEKNOLOGI REKAYASA INTERNET

JURUSAN TEKNOLOGI INFORMASI

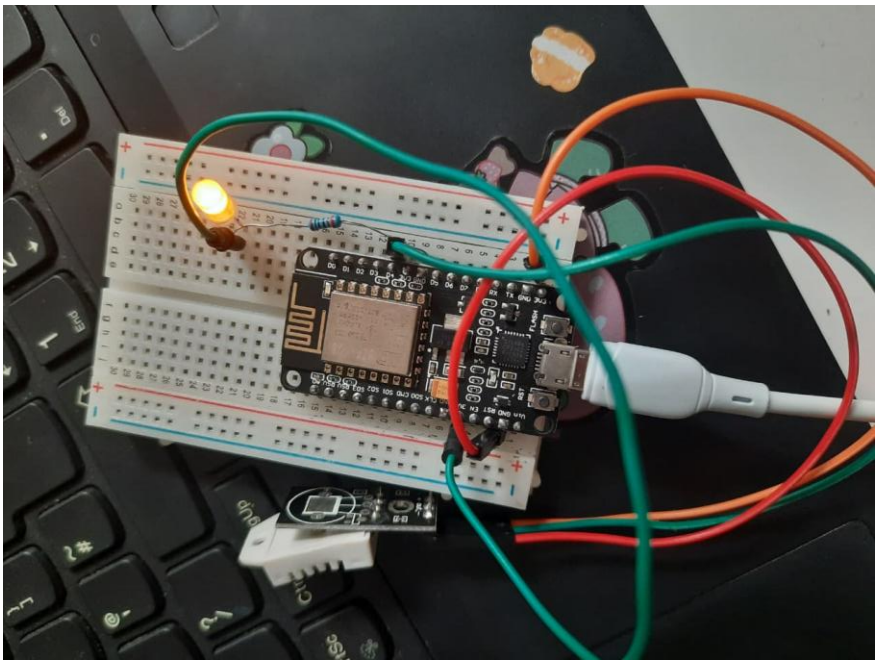
POLITEKNIK NEGERI LAMPUNG

2026/2027

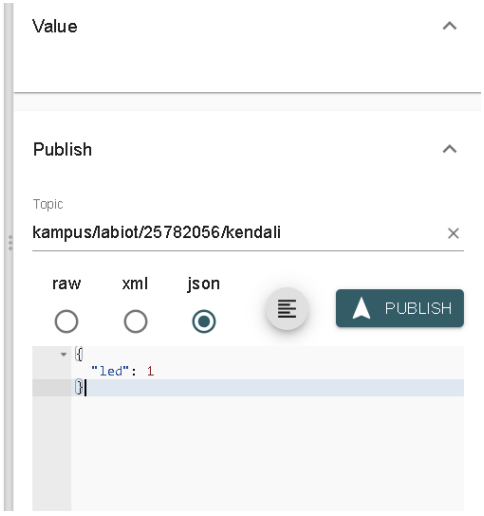
6.5.3 Praktikum 3: Sistem Terpadu Dua Arah (Full Duplex Telemetry & Control)




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MQTT Explorer:



Kesimpulan :

Komunikasi Dua Arah (*Full-Duplex*): NodeMCU ESP8266 berhasil menjalankan peran sebagai *Publisher* (mengirim data sensor & status perangkat) sekaligus *Subscriber* (menerima instruksi kendali) secara simultan tanpa *blocking*.

Umpan Balik Real-Time (*Closed-Loop Feedback*): Perubahan kondisi fisik LED saat menerima perintah kontrol langsung terekam

dan terlaporkan secara otomatis ke dalam payload telemetri berikutnya ("status_led": 1), sehingga status riil perangkat keras dapat dipantau dari jarak jauh melalui MQTT Explorer.